

On-Device Lava API (Android)

Status of this document (2026-06-02): Phases A, B, C, and D-infra of the *Lava API Android app* sub-project (spec [docs/superpowers/specs/2026-06-02-lava-api-android-app-design.md](#), plan [docs/superpowers/plans/2026-06-02-lava-api-android-app.md](#)) have landed. This document describes **only what the committed code does** — the additive SQLite storage backend (Phase A), the in-process Go embed + c-shared/JNI native build (Phase B), the `:core:apiengine` Kotlin JNI wrapper (Phase C, commit `ef109760`), and the `:api-app` headless infrastructure (Phase D-infra, commit `e62e0fa8`: the foreground `ApiEngineService`, the `ApiEngineController` state machine, the `NsdManager` advertiser, and the per-install auth-key store). The **landing UI / control screen + ViewModel** (Phase D-ui), the instrumented **Compose UI Challenge tests** (Phase E), and the **client-side distinct labelling** of Android instances (sub-project 2) are **PENDING** and are explicitly marked as such below. Nothing in this doc describes behavior that is not yet in the tree.

Classification: project-specific (the on-device-API app + Go SQLite backend are Lava-specific; the additive-backend-selector pattern is universal and is noted as such inline).

1. Motivation

The existing Lava API (`lava-api-go`) runs on a host (server, NAS, or developer workstation) backed by Postgres, advertises itself on the LAN via mDNS, and the Android client discovers and talks to it over HTTPS. The on-device API sub-project lets a **phone or tablet itself become a LAN-reachable Lava API**: the same Go server, cross-compiled and embedded in-process, backed by a local SQLite file, bound to `0.0.0.0` so other devices on the same Wi-Fi can discover and use it.

The design constraint the operator set is *additive*: "Nothing already working can be broken — only extended for more flexibility." Every existing Postgres deployment behaves byte-for-byte as before; SQLite and the embed are new, config-selected branches.

2. The additive SQLite storage backend (Phase A — landed)

2.1 Principle (universal pattern)

A **backend selector** whose default reproduces today's behavior exactly. The existing code becomes "the Postgres branch," never "the only path." Any deployment that sets `LAVA_API_PG_URL` and nothing else behaves identically to before.

Classification: the selector pattern itself is universal; the two concrete backends are Lava-specific.

2.2 Config selection

Source: lava-api-go/internal/config/config.go.

Variable	Meaning	Required when
<code>LAVA_API_STORAGE_BACKEND</code>	postgres (default) or sqlite	never (defaults to postgres)
<code>LAVA_API_PG_URL</code>	Postgres connection URL	iff backend is postgres
<code>LAVA_API_SQLITE_PATH</code>	SQLite database file path (e.g. <code>/data/lava-api.db</code> ; <code>:memory:</code> accepted)	iff backend is sqlite

`config.Load()` selects the branch in a switch `cfg.StorageBackend`:

- `postgres` → require `PGUrl` (the historical hard dependency, now scoped to this branch so existing users are unaffected).
- `sqlite` → require `SQLitePath`.
- anything else → reject with unknown storage backend `%q`.

No env var was renamed or removed; the two new variables carry placeholders in `.env.example` (§6.R). `LAVA_API_TLS_CERT/` `LAVA_API_TLS_KEY` and a valid `LAVA_API_MDNS_PORT` remain required regardless of backend.

2.3 Storage boundary and parity with Postgres

Source: [lava-api-go/internal/storage/](#).

The `storage.Storage` interface ([storage.go](#)) declares **exactly** the cache operations the handlers depend on — enumerated from the real handler consumers, nothing more:

```
type Storage interface {
    Get(ctx context.Context, key string) ([]byte, cache.Outcome,
        error)
    Set(ctx context.Context, key string, value []byte, ttl
        time.Duration) error
    Invalidate(ctx context.Context, key string) error
    Close() error
}
```

Two implementations satisfy it interchangeably:

- **postgres** — wraps the existing submodules/cache/pkg/postgres adapter (response_cache table, schema lava_api, 10-minute GC interval). Behavior is unchanged from the prior `main.go` path.
- **sqlite** — pure-Go [modernc.org/sqlite](#) (CGo-free, so it cross-compiles cleanly for Android). Behavior parity with Postgres is the explicit goal and is verified by a shared conformance harness plus a cross-backend parity test (per the plan).

`storage.New(cfg)` ([factory.go](#)) selects the implementation by `cfg.StorageBackend` and returns the `Storage` together with a `ReadinessFunc` (the /ready probe): Postgres uses `pgClient.HealthCheck`; SQLite does a trivial round-trip `Get` of a sentinel key to prove the handle answers queries.

2.4 SQLite TTL, GC, and WAL semantics

Source: [sqlite.go](#), migration [migrations/sqlite/0001_init.sql](#).

Schema mirrors the Postgres `response_cache` table with portable types:

```
CREATE TABLE IF NOT EXISTS response_cache (
    cache_key TEXT PRIMARY KEY,
    value BLOB NOT NULL,          -- BYTEA equivalent
```

```

    expires_at INTEGER -- unix NANoseconds; NULL =
        never expires
);
CREATE INDEX IF NOT EXISTS response_cache_expires_at_idx ON
    response_cache (expires_at);

```

Behavioral details the committed code implements:

- **TTL** is an absolute `expires_at` (unix nanoseconds). Set with `ttl > 0` stores `now + ttl`; `ttl <= 0` stores SQL NULL (never expires) — matching the Postgres `WHERE expires_at IS NULL OR expires_at > NOW()` contract.
- **Lazy expiry on read.** Get filters `expires_at IS NULL OR expires_at > ?`, so an expired entry reads as an `OutcomeMiss`, never a stale hit. A real DB error returns `OutcomeBypass + err` so callers fall through to the upstream — identical to `cache.Client.Get`. A stored zero-length value is a **hit** (non-nil empty slice), matching Postgres BYTEA empty-value semantics.
- **Background GC.** A goroutine runs `sweepExpired` every **10 minutes** (`sqliteGCInterval`, mirroring the Postgres GC cadence). It physically `DELETES` rows whose `expires_at` is in the past and then runs `PRAGMA incremental_vacuum` to return freed pages to the free list. `Close()` cancels the goroutine's context and joins it (no leak); `Close()` is idempotent.
- **WAL + auto_vacuum** (on-disk databases only; skipped for `:memory:`):
 - The DSN sets `journal_mode(WAL)` (concurrent readers don't block the single writer) and `busy_timeout(5000)` (ride out brief write locks).
 - `SetMaxOpenConns(1)` serializes access through one connection — the standard race-free choice for an embedded SQLite writer, and required so the connection-scoped `auto_vacuum` PRAGMA applies to the one connection every query uses.
 - `PRAGMA auto_vacuum=INCREMENTAL` followed by `VACUUM` is run **before** the migration (the `VACUUM` rewrites the DB header with the new flag; order is load-bearing — running the PRAGMA after a table exists leaves `auto_vacuum=0`).
 - After migration, `PRAGMA journal_mode` is re-queried and the constructor **fails loudly** if WAL did not engage, refusing to run in a weaker (DELETE) journal mode that would drop the concurrent-reader guarantee.

3. The in-process Go embed (Phase B — landed)

Source: [lava-api-go/internal/mobile/mobile.go](#) and [tls.go](#).

`internal/mobile` is the small, JNI-friendly lifecycle surface that boots the **full production API in-process** inside the Android app (Phases C/D will call it). Three exported functions, designed so only string/error cross the boundary:

```
func Start(configJSON string) error // parse config → SQLite +
    full router + TLS, listen, return when accepting
func Stop() error // graceful shutdown + close
    storage; errors if nothing running
func Status() string // JSON: state, scheme,
    bindAddr, port, requestCount, backend, version, auth*
```

Start's configJSON shape:

```
{ "bindAddr": "0.0.0.0", "port": 8443, "sqlitePath": "/data/
x.db", "authSharedKey": "<base64-uuid>", "authFieldName": "<header>" }
```

— `bindAddr/port` optional (default `0.0.0.0:8443`); `sqlitePath` required; `authSharedKey/authFieldName` optional (see §3.4). Exactly one server instance may run per process (a package-level handle guarded by a mutex; Start while running returns an error).

3.1 Why c-shared, not gomobile

Source: [lava-api-go/cmd/lavaapi-cshared/main.go](#).

`gomobile bind` is **BLOCKED** on this module. `lava-api-go`'s `go.mod` uses relative replace `../submodules/*` directives for the 16 vasic-digital submodules; `gomobile`'s overlay-module generator cannot resolve them (it writes a 0-byte `go.mod`; proven on NDK 21.4 + 25.1). The chosen replacement is **go build -buildmode=c-shared**, which uses ordinary Go module resolution and therefore HONORS the replace directives. It is still an in-process, cross-compiled Go artifact.

The tradeoff: `c-shared` exports a flat C ABI, not a generated Java class hierarchy. A **hand-written JNI bridge** adapts the four exported C functions to the JNI naming convention the Kotlin side will bind.

The exported C surface (all strings NUL-terminated UTF-8; every returned `char*` is heap-allocated by `C.CString` and the caller **MUST** release it via `LavaApiFree`):

```
extern char* LavaApiStart(char* configJSON); // "" on success,
      else error msg
extern char* LavaApiStop(void); // "" on success,
      else error msg
extern char* LavaApiStatus(void); // status JSON
      document
extern void LavaApiFree(char* p); // free a returned
      string
```

3.2 The JNI contract

Source: [lava-api-go/cmd/lavaapi-cshared/jni/jni_bridge.c](#) and [CMakeLists.txt](#).

`jni_bridge.c` builds `liblavaapi_jni.so`, which links against the prebuilt `liblavaapi.so` (from the c-shared build) and exposes the entry points the Kotlin side will bind:

```
package digital.vasic.lava.apigo
```

```
object LavaNative {
    external fun nativeStart(configJson: String):
        String // "" on success, else error message
    external fun nativeStop(): String // ""
        on success, else error message
    external fun nativeStatus(): String //
        status JSON document
}
```

JNI symbol names (package dots → underscores, then class, then method):

- `Java_digital_vasic_lava_apigo_LavaNative_nativeStart`
- `Java_digital_vasic_lava_apigo_LavaNative_nativeStop`
- `Java_digital_vasic_lava_apigo_LavaNative_nativeStatus`

A Kotlin object compiles to a final class with a static `INSTANCE`; its external functions register as native methods that receive a `jclass`, so the bridge functions take `(JNIEnv*, jclass, ...)`. The bridge copies each Go-returned C string into a JVM string via `NewStringUTF` and immediately releases the C buffer via `LavaApiFree` so nothing leaks across the boundary.

`jni_bridge.c` and its `CMakeLists.txt` live in the `jni/` **subdirectory**, NOT the `cgo` main package, because `jni_bridge.c` `#includes` `liblavaapi.h` — the *cgo-generated* header the c-shared build PRODUCES — so it cannot be compiled by the `cgo` build itself; it is compiled later by the Android NDK toolchain via `externalNativeBuild { cmake { ... } }`.

LANDED (Phase C, commit ef109760): the Kotlin LavaNative object, the `:core:apiengine` wrapper module, and the externalNativeBuild Gradle wiring that loads these native libraries now exist in the tree and match the contract above exactly. See [§4A — Phase C](#) for the full mapping.

3.3 What the embed serves (full production router) + TLS

The embed builds the **EXACT same Gin engine** the production `cmd/lava-api-go` binary builds, via the shared `internal/router.Build` constructor ([router.go](#)) — `search / browse / topic / forum / torrent / login / favorites / captcha, /v1/{provider}/...`, plus `/health` and `/ready`. The provider registry wires the same adapters production wires (`rutracker`, `nmclub`, `kinozal`, `archive.org`, `gutenberg`). Sharing one router constructor is deliberate (§6.J): a divergent embed router would be a bluff vector — green tests against an embed serving different routes than production would guarantee nothing about production.

The **only** transport difference from the LAN binary: the embed serves HTTP/1.1 + HTTP/2 over TLS using a standard `net/http` server; the binary additionally serves HTTP/3 (QUIC). HTTP/3 is not required for the embed — LAN clients that discover the on-device API speak ordinary HTTPS. The embed explicitly disables Brotli / protocol-metric / AltSvc middlewares (transport tuning, not security) so repeated in-process Start/Stop cycles don't touch the Prometheus default registerer.

TLS ([tls.go](#)):

- A self-signed **ECDSA P-256** cert + key is generated on first boot and persisted next to the SQLite DB (`lava-embed-cert.pem / lava-embed-key.pem`; the key is written 0600). Validity is 10 years (rotation handling is a later sub-project).
- If both files already exist and parse, they are **reused** across restarts, so a peer that pinned the leaf cert out-of-band does not see it change every boot.
- **IP SANs:** loopback (IPv4 + IPv6) plus the host's non-loopback, non-link-local LAN IPs discovered at boot (`localIPs()`), so a peer addressing the server by LAN IP does not hit a host-mismatch error.
- **No wildcard DNS SAN.** DNS SANs are limited to `localhost` — a bare `*` wildcard SAN matched ANY hostname and defeated name verification (a security-review finding; fixed in commit 6f751495). Peers address the embed by LAN IP (covered by the IP SANs) or by `localhost` on-device. Out-of-band leaf trust on the consuming

devices is the model for now (full client-trust / local-CA handling is a later sub-project).

3.4 The Lava-Auth gate on the embed (same mechanism as the host)

Because the embed is network-exposed (§3.5), it **MUST authenticate** — there is no "trusted LAN" shortcut. It enforces the **identical** Lava-Auth gate the production LAN binary uses (`internal/auth.NewMiddleware + NewBackoffMiddleware`, source [middleware.go](#)):

- The client presents a credential in a configurable header (default Lava-Auth). The credential is base64(16-byte UUID blob).
- The middleware base64-decodes the header, computes `hex(HMAC-SHA256(uuid_bytes, secret))`, and compares (in constant time, to defeat timing side-channels) against the active-clients allowlist.
- Active hash → request served (`client_name` set, backoff reset).
Retired hash → 426 Upgrade Required. Missing / malformed / unknown → 401 and the per-IP backoff ladder advances; repeated failures escalate to 429 on the default ladder 2s, 5s, 10s, 30s, 1m, 1h.
- `/health` and `/ready` are registered **before** the auth middleware so the orchestrator probes work without a credential.

The embed's app-managed key flow:

- The host app supplies the credential via `authSharedKey` in the Start config.
- **If none is supplied, the embed GENERATES one** (random 16-byte UUID, base64-encoded) so the gate is never disabled. The generated/in-effect key is surfaced via `Status()` (the `authKey` field) — and **only** there — so the app can display it for pairing. It is deliberately never written to a log line (§6.H / §6.AC redaction). The plaintext blob is zeroized after hashing.
- `Status().authEnabled` is always `true` while running.

LANDED (Phase D-infra, commit e62e0fa8): persisting the key across app restarts is now the `:api-app ApiKeyStore`'s responsibility (a per-install base64-UUID blob held in `EncryptedSharedPreferences`); see [§4B.4 — auth-key store](#). The UI that shows it for pairing remains **PENDING (Phase D-ui)**. The Go embed surface here only **ENFORCES** whatever key is in effect and reports it via `Status()`.

3.5 Network exposure (0.0.0.0) and the trust boundary

The embed binds `0.0.0.0` by default — network exposure is the intended design. Loopback-only would defeat the feature (the whole point is that a phone running the embed becomes a LAN-reachable Lava API for tablets, TVs, and other phones). The bind address is validated with `net.ParseIP`; non-loopback (including the `0.0.0.0` wildcard) is explicitly allowed, and only unparseable IPs are rejected. The automated security-review suggestion of loopback-only was therefore explicitly overridden in the code, with the Lava-Auth gate (§3.4) as the compensating control: anyone on the LAN can *reach* the port, but only a holder of the auth key gets a non-401 response.

3.6 mDNS identity (Android instance vs host instance)

The embed is honestly the Go engine on Android with SQLite, so it will advertise:

- Service type `_lava-api._tcp` (release) / `_lava-api-dev._tcp` (debug) — the same types the client already watches (catalog in `core/data/.../DiscoveryServiceTypes.kt`: `SERVICE_TYPE_GO = "_lava-api._tcp"`, `SERVICE_TYPE_GO_DEV = "_lava-api-dev._tcp"`, `SERVICE_TYPE_KTOR = "_lava._tcp"`).
- TXT: `engine=go`, **`platform=android`**, `storage=sqlite`, `version=<n>`.

The existing host instance keeps advertising `engine=go` with `platform=server` (or absent) — unchanged; both appear in the client's list. The "distinct identity" is realized via the `platform` TXT key, which the client (sub-project 2) will label distinctly without any discovery re-architecture. Instances without `platform` render exactly as today (backward compatible).

LANDED (Phase D-infra, commit `e62e0fa8`): the on-device `NsdManager` *registration* (the embed advertising itself) now exists in `:api-app` as `NsdMdnsAdvertiser`; see [§4B.3 — mDNS advertiser](#). The client-side *discovery* that consumes these TXT records already exists in `LocalNetworkDiscoveryServiceImpl` and is described in `LOCAL_NETWORK_DISCOVERY.md`.

4A. Phase C — the `:core:apiengine` JNI wrapper (landed)

Source: [core/apiengine/](#) (commit ef109760).

`:core:apiengine` is an Android library that exposes the embedded lava-api-go server (the Go c-shared library + JNI bridge from Phase B) behind an ordinary Kotlin suspend-function API. Consumers (`:api-app`, future wiring) depend ONLY on this module's public surface — never on LavaNative directly. Per the Decoupled Reusable Architecture rule this is Lava-domain glue: it introduces no logic another vasic-digital project would consume.

4A.1 The public Kotlin surface (`ApiEngine`)

Source: [ApiEngine.kt](#) (package `lava.apiengine`).

```
interface ApiEngine {
    suspend fun start(config: ApiConfig): Result<ApiStatus> //
        success → running ApiStatus; failure → native error
    suspend fun stop(): Result<Unit> //
        failure when nothing is running (not a silent no-op)
    fun status(): ApiStatus //
        non-suspending snapshot
}
```

`ApiConfig` is the start-time configuration, serialized to the exact JSON shape `internal/mobile.startConfig` parses:

Field	Type	Default	Meaning
<code>bindAddr</code>	<code>String</code>	<code>"0.0.0.0"</code>	bind IP; wilddcard so LAN peers can reach the API (loopback-only would defeat the feature)
<code>port</code>	<code>Int</code>	<code>8443</code>	TCP listener port, matching the production LAN listener
<code>sqlitePath</code>	<code>String</code>	<i>(required)</i>	absolute path to the SQLite DB file; TLS material is persisted beside it
<code>authSharedKey</code>	<code>String?</code>	<code>null</code>	the base64-UUID credential; when null the embed GENERATES one and surfaces it via <code>ApiStatus.authKey</code> (there is no unauthenticated mode)
<code>authFieldName</code>	<code>String</code>		

Field	Type	Default	Meaning
		"Lava-Auth"	HTTP header the credential is read from

`ApiStatus` mirrors the `JSON internal/mobile.Status()` returns: `state` ("running"/"stopped"), `bindAddr`, `port`, `requestCount`, `backend` (always "sqlite"), `version`, `scheme` (always "https"), `authEnabled` (always true while running), `authFieldName`, and the nullable `authKey` (the in-effect credential the host app can display for pairing; null when stopped).

Exactly one server instance runs per process (enforced by the native side): `start` while already running returns a failed `Result`.

4A.2 NativeApiEngine — the real JNI-backed implementation

Source: [NativeApiEngine.kt](#).

`NativeApiEngine` is the production `ApiEngine`. It maps the Kotlin surface onto the three `LavaNative` external functions:

- `start(config)` serializes `config` to JSON via `kotlinx.serialization` (`ConfigDto` — field names `bindAddr/port/sqlitePath/authSharedKey/authFieldName` matching the Go `startConfig` tags; a null `authSharedKey` is emitted as "", which the Go side treats as "generate one"), calls `LavaNative.nativeStart(configJson)`, and — because the native `Start/Stop` contract is "empty string on success, error message otherwise" — maps a non-empty return to `Result.failure(ApiEngineException(err))`. On success it parses `LavaNative.nativeStatus()` into `ApiStatus`.
- `stop()` calls `LavaNative.nativeStop()` and applies the same empty-string = success mapping.
- `status()` parses `LavaNative.nativeStatus()` (the `StatusDto` fields are omitted on the Go side when stopped, so they default to stopped-state zero values).

`start` and `stop` run on an injectable `CoroutineDispatcher` (default `Dispatchers.IO`) because the native calls may block on socket bind / TLS material generation / SQLite open. The class requires a real Android device/emulator (the native .so is Android-only), so it is NOT JVM-unit-testable; its behavioral contract is verified by the Phase E Compose UI Challenge against a real emulator (PENDING). The JVM-testable contract lives in `FakeApiEngine`.

4A.3 LavaNative — the JNI binding

Source: [LavaNative.kt](#) (package `digital.vasic.lava.apigo` — chosen to match the JNI symbol encoding).

```
package digital.vasic.lava.apigo

internal object LavaNative {
    init { System.loadLibrary("lavaapi_jni") }
    external fun nativeStart(configJson: String):
        String // "" on success, else error message
    external fun nativeStop(): String // ""
        on success, else error message
    external fun nativeStatus(): String //
        status JSON document
}
```

The package + object + method names MUST match the JNI symbol encoding in `jni_bridge.c` exactly (e.g. `Java_digital_vasic_lava_apigo_LavaNative_nativeStart`). `System.loadLibrary("lavaapi_jni")` resolves to `liblavaapi_jni.so`; its transitive dependency on `liblavaapi.so` (the prebuilt Go c-shared library) is satisfied because BOTH libraries are packaged into the same `jniLibs/<abi>/` directory of the APK. The object is internal — consumers go through `NativeApiEngine`, never here.

4A.4 FakeApiEngine — the behaviorally-equivalent JVM fake

Source: [FakeApiEngine.kt](#).

Per the Anti-Bluff Pact Third Law, the fake reproduces the production / Go BRANCHES that affect callers — it is NOT a "simpler than reality" stub:

- **Single instance.** `start` while already running fails with an "already running" error matching the Go "mobile: server already running ...".
- **Stop is not a no-op.** `stop` without a running server fails with a "no server running" error matching `internal/mobile.Stop`.
- **Status reflects last start/stop.** stopped before any start; running with the started config's `bindAddr/port` after start; reverts to stopped after stop.
- **Auth always on while running.** `start` surfaces an auth key (the config's `authSharedKey` when provided, else a deterministic generated stand-in) and reports `authEnabled == true` — no unauthenticated mode.

- **Error injection.** `failWith(error)` makes the next start/stop propagate a `Result.failure`, exercising callers' error paths.
`recordRequests(count)` simulates inbound traffic so `status()` reports a non-zero count.

4A.5 The native build pipeline (`buildCshared` → `jniLibs` → `externalNativeBuild`)

Source: `core/apiengine/build.gradle.kts`.

Two native libraries are packaged per ABI into this module's AAR, for the three ABIs `arm64-v8a` / `x86_64` / `armeabi-v7a` (NDK 25.1.8937393, CMake 3.22.1):

1. **`liblavaapi.so`** — the prebuilt Go c-shared library. A `buildCshared` Gradle Exec task shells out to `lava-api-go/scripts/build-cshared.sh` for the module's ABIs, writing `lava-api-go/build/jniLibs/<abi>/liblavaapi.{so,h}`. The task declares those `.so/.h` as outputs so it is **up-to-date when all expected outputs are already present** — a clean assemble does not force a multi-minute Go rebuild when the artifacts are fresh. On a Windows build host the task fails honestly (the Go c-shared/JNI path is unsupported there) rather than producing a broken artifact.
`sourceSets.main.jniLibs.srcDir(prebuiltJniLibsDir)` points AGP at that output dir so it packages `liblavaapi.so` per ABI directly (Go is NOT recompiled from Gradle — the script output is LOCATED and staged).
2. **`liblavaapi_jni.so`** — the hand-written JNI bridge. AGP's `externalNativeBuild { cmake { path = jniCmakeLists } }` points at the canonical `lava-api-go/cmd/lavaapi-cshared/jni/CMakeLists.txt`, which imports `liblavaapi.so` as an IMPORTED SHARED lib and links the bridge against it.
`-DLAVAAPI_PREBUILT_DIR=<abs path>` is passed as a CMake argument so the `CMakeLists` finds the prebuilt `.so/.h` (laid out as `<ABI>/liblavaapi.{so,h}`).

The native-build / `jniLibs-merge` tasks (`externalNativeBuild*`, `configureCMake*`, `buildCMake*`, `mergeJniLibFolders`) all `dependsOn(buildCshared)` so the prebuilt inputs exist before CMake configures and before the `jniLibs` merge reads the directory.

4B. Phase D-infra — the :api-app module (landed)

Source: [api-app/](#) (commit e62e0fa8).

:api-app is the standalone *Lava API* Android application. It hosts the embedded lava-api-go server (via :core:apiengine) as an on-device, LAN-reachable HTTPS API, advertised over mDNS so other devices discover it. Phase D-infra delivers the **headless infrastructure** only: the foreground Service, the Android-free lifecycle controller, the mDNS advertiser, and the per-install auth-key store. The Compose landing UI, the refined notification copy, and the ViewModel are **Phase D-ui (PENDING)**; this module ships a placeholder MainActivity (its onCreate calls `setContent { }`) so the manifest is valid and the APK assembles.

4B.1 Module identity, signing, and permissions

Source: [api-app/build.gradle.kts](#), [AndroidManifest.xml](#).

- **Application ID** `digital.vasic.lava.api`; the debug build applies `applicationIdSuffix = ".dev"` → `digital.vasic.lava.api.dev`, so the release and dev variants install side-by-side and advertise distinct mDNS service types (§4B.3). `versionCode = 1`, `versionName = "0.1.0"`, `minSdk = 23` (the per-install auth-key store relies on `EncryptedSharedPreferences`, which requires API 23+; the client :app keeps `minSdk 21`).
- **Shared signing.** The signing block mirrors :app's .env-driven inputs verbatim — the SAME `KEYSTORE_PASSWORD` + `KEYSTORE_ROOT_DIR` keystore material (debug → `keystores/debug.keystore` alias `debug`; release → `keystores/release.keystore` alias `release`). No parallel signing scheme is invented (§6.R / §6.H). Release uses `postprocessing (remove-unused-code + remove-unused-resources + optimize, isObfuscate = false)`.
- **Permissions** the embed needs while serving: `INTERNET`, `ACCESS_NETWORK_STATE`, `ACCESS_WIFI_STATE`, `CHANGE_WIFI_MULTICAST_STATE` (mDNS multicast), `FOREGROUND_SERVICE` + `FOREGROUND_SERVICE_DATA_SYNC`, `POST_NOTIFICATIONS`, and `WAKE_LOCK`. The application sets `android:usesCleartextTraffic="true"` (matching the client :app). The Service is declared `android:exported="false"` with `android:foregroundServiceType="dataSync"` (the closest standard type for a process that continuously serves LAN requests + advertises over mDNS). The `ApiApplication` is a `@HiltAndroidApp`

root; D-infra wires no Hilt modules (the Service constructs its controller directly) — the annotation is present so D-ui can add injected ViewModels without re-shaping the app.

4B.2 The foreground `ApiEngineService` + OS locks

Source: [service/ApiEngineService.kt](#).

A foreground Service that owns the Android-free `ApiEngineController` and drives it via intent actions:

- **Intent actions:** `ACTION_START` (`lava.api.app.action.START`, the default), `ACTION_STOP`, `ACTION_RESTART`. `onStartCommand` promotes to foreground *immediately* (`startForeground` with a baseline notification) so the OS does not kill the process before the first state emission, then dispatches the matching `controller.start()` / `stop()` / `restart()` on a service-scoped `CoroutineScope` (cancelled in `onDestroy`). It returns `START_STICKY`.
- **OS locks held while serving** (acquired on `Running`, released on `Stopped` / `Error` / `onDestroy`):
 - a Wi-Fi **MulticastLock** so mDNS multicast reaches the LAN;
 - a Wi-Fi **WifiLock** in `WIFI_MODE_FULL_HIGH_PERF` so the radio stays up;
 - a **partial WakeLock** (`PARTIAL_WAKE_LOCK`) so the CPU keeps serving while the screen is off. All three are `setReferenceCounted(false)` and released the moment the API stops (idempotent release guarded by `isHeld`).
- **The controller is built** with a `NativeApiEngine`, an `NsdMdnsAdvertiser` whose engine identity is `G0_DEV` when the package name ends in `.dev` else `G0`, an `ApiKeyStoreImpl.create(applicationContext)`, a `NetworkInterfaceLanIpProvider`, and a `sqlitePath` of `filesDir/lava-api.db`.
- **Foreground notification:** an ongoing, low-importance notification on channel `lava.api.app.server`. When `Running` the body shows "`<url> • <n> req`" (reachable URL + live request count); it carries **Stop + Restart** actions (each a `PendingIntent.getService` with the matching action) and a content-intent back to `MainActivity`. The copy is intentionally minimal (Phase D-ui refines it).

4B.2a The `ApiEngineController` state machine

Source: [control/ApiEngineController.kt](#), [control/ApiControlState.kt](#).

The controller is the **Android-free lifecycle core** — it has NO Service / Context dependency, so it is unit-tested with the real controller wired to FakeApiEngine + fakes for the advertiser/keyStore (the load-bearing Anti-Bluff test ApiEngineControllerTest). It owns the ApiEngine, the MdnsAdvertiser, and the ApiKeyStore, and exposes state: StateFlow<ApiControlState> as the single source of truth the Service + UI observe.

The ApiControlState sealed interface forms the cycle Stopped → Starting → Running → Stopping → Stopped, with Error reachable from Starting/Stopping:

State	Meaning
Stopped	no server running (initial state)
Starting	a start was requested; the engine has not yet reported Running
Running	the embed is up; carries url, lanIps, port, requestCount, authKey, authFieldName — all from the real post-start ApiStatus + the host's discovered LAN IPs, never synthesized
Stopping	a stop was requested; shutdown not yet confirmed
Error(message)	the last start/stop failed; message is the engine's error text

Behavioral details the committed code implements:

- **start()** emits Starting, fetches the persisted key via `keyStore.getOrCreate()`, builds an `ApiConfig` (port, `sqlitePath`, the key as `authSharedKey`, `keyStore.fieldName` as `authFieldName`), and calls `engine.start(config)`. On success it calls `onStarted(status)`: it resolves the host's LAN IPs (`lanIpProvider()`), **registers the mDNS advertisement on the running port**, and emits Running with the reachable URL built as `<scheme>://<first LAN IP or 127.0.0.1>:<port>`. On engine failure it emits Error and does **NOT** advertise.
- **stop()** emits Stopping, **unregisters the advertisement first** (so a stale record never points at a dead listener), then calls `engine.stop()`: success → Stopped, failure → Error.
- **restart()** is `stop()` then `start()`; if the stop surfaced an Error it halts there (a failed stop leaves the embed in an unknown state; surfacing Error is the honest behavior).
- The default port is read from `ApiConfig`'s own default so the literal lives in `:core:apiengine`, not duplicated here (§6.R). The host's LAN

IPs come from `NetworkInterfaceLanIpProvider` ([net/LanIpProvider.kt](#)), which enumerates the real non-loopback, non-link-local IPv4 interface addresses — not a synthesized value (Sixth-Law clause 3: the URL the UI shows and the cert SANs the embed generates both depend on these being the TRUE reachable IPs).

4B.3 The mDNS advertiser (`NsdMdnsAdvertiser`)

Source: [service/MdnsAdvertiser.kt](#), [service/NsdMdnsAdvertiser.kt](#).

`MdnsAdvertiser` is a two-method interface (`register(port) / unregister()`); `NsdMdnsAdvertiser` is the `NsdManager`-backed implementation. It registers a single `NsdServiceInfo` named "Lava API" for the build-appropriate service type, on the running port, with the TXT records the pure `buildTxtRecords` builder produces. `register` first calls `unregister()` (a previous registration is torn down before re-registering); `unregister` is safe to call when not registered.

- **Service types** (matching `DiscoveryServiceTypeCatalog` in `core/data` — the cross-process mDNS protocol contract):
 - release build → `_lava-api._tcp` (engine identity `GO`)
 - debug build → `_lava-api-dev._tcp` (engine identity `GO_DEV`)
- **TXT records** (the pure, unit-testable wire contract via `buildTxtRecords` — verified by `MdnsAdvertiserTxtTest` without touching Android): `engine=go` (or `go-dev`), `platform=android`, `storage=sqlite`, `version=<n>`. The version is read from a `versionProvider` lambda (`engine.status().version`) at register time so the advertised version tracks the actual running embed, not a hardcoded literal (§6.R).

The service-type + TXT literals are kept in `:api-app` (rather than depending on `:core:data`) so the API app does not pull in the entire client data layer; they are identical to `DiscoveryServiceTypeCatalog.SERVICE_TYPE_GO / SERVICE_TYPE_GO_DEV`.

4B.4 The per-install auth-key store (`ApiKeyStore`)

Source: [auth/ApiKeyStore.kt](#).

`ApiKeyStore` is a two-member interface — `getOrCreate(): String` and `val fieldName: String`. `ApiKeyStoreImpl` implements it:

- The credential is the **production wire shape the Go embed accepts**: the base64 (standard alphabet, padded, single line)

encoding of a random 16-byte UUID blob. This is EXACTLY what `internal/mobile.generateAuthKey` mints and what `internal/auth.NewMiddleware` base64-decodes before HMAC-ing — so a key produced here is honoured by the embed when passed as `ApiConfig.authSharedKey`.

- **First run** generates a fresh key with `SecureRandom` (16 bytes, `UUID_LEN`), base64-encodes it (`android.util.Base64 NO_WRAP`, matching Go's `base64.StdEncoding`), and persists it; every subsequent call (this run or a later install lifetime) returns the SAME key so paired peer devices keep working. `getOrCreate` is `@Synchronized`.
- **Encrypted at rest:** `ApiKeyStoreImpl.create(context)` builds an `EncryptedSharedPreferences` (AES256-SIV key encryption + AES256-GCM value encryption, AES256_GCM master key) in the `lava_api_auth` prefs file. Pure- JVM unit tests construct `ApiKeyStoreImpl` directly with an in-memory `SharedPreferences` fake + a `java.util.Base64-backed KeyEncoder` that yields the identical alphabet (the encoder is injectable for exactly this reason).
- **\$6.H credential hygiene:** the key is NEVER written to a log line, never concatenated into an exception message, never returned from `toString`. The plaintext blob is `fill(0)`'d in a `finally` block after encoding; the base64 String leaves the process only through the deliberate `display/config` path (D-ui). The default `fieldName` is "Lava-Auth", matching the embed.

PENDING (Phase D-ui): the landing screen / control screen, the `ViewModel`, and the refined notification copy that DISPLAY the URL, the auth key (for pairing), and the live request count are not yet in the tree — `MainActivity` is a placeholder. **PENDING (Phase E):** the instrumented Compose UI Challenge tests that boot the embed on a real emulator and issue a real HTTPS request.

PENDING (sub-project 2): the client-side distinct labelling of `platform=android` instances in the discovery list.

4. Diagrams

4.1 Component / module diagram (what has landed)

flowchart TB

```
    subgraph LavaApiGo["lava-api-go (Go)"]
        cfg["internal/config<br/>storage selector<br/>postgres |
sqlite"]
        router["internal/router.Build<br/>FULL production Gin
engine<br/>(shared by binary + embed)"]
    end
```

```

        auth["internal/auth<br/>Lava-Auth gate<br/>HMAC-SHA256 +
backoff ladder"]
        subgraph Storage["internal/storage"]
            iface["Storage interface<br/>Get / Set / Invalidate /
Close"]
            pg["postgres impl<br/>(submodules/cache)"]
            sq["sqlite impl<br/>modernnc.org/sqlite<br/>WAL +
INCREMENTAL auto_vacuum + GC"]
        end
        mobile["internal/mobile<br/>Start / Stop / Status<br/
>0.0.0.0 + self-signed TLS"]
        cshared["cmd/lavaapi-cshared<br/>C ABI: LavaApiStart/Stop/
Status/Free"]
    end

    subgraph NativeBuild["c-shared build (build-cshared.sh)"]
        so["liblavaapi.so + liblavaapi.h<br/>per ABI: arm64-v8a /
x86_64 / armeabi-v7a"]
        jni["jni_bridge.c + CMakeLists.txt<br/>liblavaapi_jni.so"]
    end

    subgraph Android["Android :api-app + :core:apiengine"]
        kotlin["LavaNative (external fun)<br/>(Phase C – landed)"]
        engine[":core:apiengine<br/>ApiEngine /
NativeApiEngine<br/>(Phase C – landed)"]
        ctrl["ApiEngineController<br/>Stopped/Starting/Running/
Stopping/Error<br/>(Phase D-infra – landed)"]
        svc["ApiEngineService (foreground)<br/>Wifi/Multicast/Wake
locks<br/>(Phase D-infra – landed)"]
        adv["NsdMdnsAdvertiser<br/>_lava-api(-dev)._tcp + TXT<br/
>(Phase D-infra – landed)"]
        keys["ApiKeyStore<br/>EncryptedSharedPreferences<br/
>(Phase D-infra – landed)"]
        ui["ApiControl landing UI + ViewModel<br/>(Phase D-ui –
PENDING)"]
    end

    cfg --> Storage
    iface --> pg
    iface --> sq
    mobile --> router
    router --> auth
    router --> iface
    mobile --> sq
    cshared --> mobile
    cshared --> so
    so --.includes.--> jni
    jni -- JNI --> kotlin
    kotlin --> engine
    svc --> ctrl
    ctrl --> engine
    ctrl --> adv

```

```
ctrl --> keys
svc --> ui
```

```
classDef pending fill:#eee,stroke:#999,stroke-dasharray: 5 5;
class ui pending;
```

4.2 Start / Stop / Restart lifecycle state machine

```
stateDiagram-v2
    [*] --> Stopped
    Stopped --> Starting: Start(configJSON)
    Starting --> Running: listener accepting<br/>(net.Listen
returned)
    Starting --> Error: bind / storage / TLS failure
    Running --> Stopping: Stop()
    Stopping --> Stopped: graceful shutdown + store.Close()
    Running --> Running: Start() while running → error<br/>
>(exactly one instance per process)
    Error --> Starting: Start(configJSON) retry
    Stopped --> Stopped: Stop() → error "no server running"

    note right of Running
        Status() reports:
        state=running, scheme=https,
        bindAddr, port, requestCount,
        backend=sqlite, version,
        authEnabled=true, authFieldName, authKey
    end note
```

The Go embed itself exposes only Stopped and Running via `Status().state`. The Starting/Stopping/Error transient states are now realized in Kotlin by the `:api-app ApiEngineController` (Phase D-infra — landed; see §4B.2a): the controller emits Starting before `engine.start()` and Stopping before `engine.stop()`, and Error(message) when either fails. The UI that RENDERS these states is PENDING Phase D-ui.

4.3 mDNS discovery + authenticated request sequence

```
sequenceDiagram
    participant P as Phone running :api-app<br/>(ApiEngineService
+ controller – landed)
    participant N as LAN (mDNS / multicast)
    participant C as Other Lava client

    P->>P: Service.start → controller.start → engine.start<br/>
>(0.0.0.0:8443, SQLite, self-signed TLS)
    P->>P: ApiKeyStore.getOrCreate → Lava-Auth key (persisted,
EncryptedSharedPreferences)
```

```

    Note over P,N: Phase D-infra – NsdMdnsAdvertiser.register
(NsdManager.registerService)
    P->>N: advertise _lava-api._tcp<br/>TXT engine=go,
platform=android, storage=sqlite, version=n
    C->>N: NsdManager discover _lava-api._tcp (already in-tree)
    N->>C: resolved host:port + TXT
    C->>C: map to Endpoint.GoApi(host, port)<br/>label
platform=android distinctly (PENDING sub-project 2)
    C->>P: HTTPS GET /health (no auth required)
    P->>C: 200 OK
    C->>P: HTTPS GET /v1/{provider}/search?... (Lava-Auth: <key>)
    P->>P: auth middleware: HMAC-SHA256 vs allowlist (constant-
time)
    alt valid key
        P->>C: 200 + real JSON
    else missing / wrong key
        P->>C: 401; backoff ladder advances → 429 on repeat
    end

```

4.4 Build pipeline (Go → per-ABI .so → jniLibs → app)

flowchart LR

```

    src["internal/mobile +<br/>cmd/lavaapi-cshared"]
    subgraph build["build-cshared.sh (per ABI)"]
        cc["CGO_ENABLED=1<br/>GOOS=android GOARCH=&lt;arch&gt;<br/>
>CC=NDK clang"]
        gob["go build -buildmode=c-shared"]
        verify["llvm-nm -D → assert LavaApiStart present<br/>+
sha256 + size"]
    end
    out["build/jniLibs/&lt;abi&gt;/<br/>liblavaapi.{so,h}<br/>
>(gitignored)"]
    task["core:apiengine buildCshared task<br/>(invokes build-
cshared.sh;<br/>up-to-date when outputs fresh)"]
    cmake["AGP externalNativeBuild → NDK CMake<br/>jni_bridge.c →
liblavaapi_jni.so<br/>(Phase C – landed)"]
    jnilibs["core:apiengine jniLibs srcDir +<br/>
>externalNativeBuild output<br/>liblavaapi.so + liblavaapi_jni.so
per ABI<br/>(Phase C – landed)"]
    app[":api-app APK (depends on :core:apiengine)<br/>(Phase D-
infra – landed)"]

    src --> cc --> gob --> verify --> out
    task --> out
    out --> cmake --> jnilibs --> app

```

5. Building the native libraries

See the script guide [docs/scripts/build-cshared.sh.md](#) for the authoritative reference. In brief:

```
cd lava-api-go
./scripts/build-cshared.sh           # all three ABIs
./scripts/build-cshared.sh arm64-v8a # just the critical ABI
```

ABI	GOARCH	NDK clang wrapper (API=28 default)
arm64-v8a	arm64	aarch64-linux-android28-clang
x86_64	amd64	x86_64-linux-android28-clang
armeabi-v7a	arm	armv7a-linux-androideabi28-clang

Outputs land in `lava-api-go/build/jniLibs/<abi>/liblavaapi.{so,h}` (gitignored — only the source is committed). The script prints each `.so`'s size + sha256 and asserts `LavaApiStart` is in the dynamic symbol table via `llvm-nm`. A missing NDK toolchain or NDK clang is reported as a hard failure (exit 2 for config errors, 1 for build/symbol failures) — no `.so` is faked (§6.J / Anti-Bluff).

LANDED (Phases C + D-infra): the `:core:apiengine` wrapper (Phase C, commit `ef109760`) is now the Gradle consumer of these native libraries — its `buildCshared` task invokes `build-cshared.sh` and its `externalNativeBuild` packages `liblavaapi.so` + `liblavaapi_jni.so` per ABI (see §4A.5). The `:api-app` module (Phase D-infra, commit `e62e0fa8`) depends on `:core:apiengine` and ships the foreground service + controller + on-device mDNS advertiser + auth-key store (see §4B).

PENDING (Phases D-ui / E): the landing UI + control screen + ViewModel (Phase D-ui) and the instrumented Compose UI Challenge tests that boot the embed on a real emulator and issue a real HTTPS request (Phase E) are not yet in the tree.

5A. API↔embed sync guarantee (no drift) — §11.4.69 / §6.J

The on-device embed (`liblavaapi.so`) MUST contain EXACTLY the current lava-api-go source codebase — a stale `.so` would serve old API behaviour while every other gate looked green (the canonical "tests pass, wrong-version feature" bluff). A four-part mechanism, anchored on ONE hash function, makes drift mechanically impossible to hide:

1. **Single source of truth** — `scripts/compute-api-source-hash.sh`. A deterministic bare-64-hex sha256 over the EXACT file set that compiles into the `.so`: every non-test `.go` under `lava-api-go/cmd/` `lavaapi-cshared` + `lava-api-go/internal`, plus `go.mod` + `go.sum`. The list is byte-sorted and each file's relative path + content is hashed, so a rename, move, content edit, or dependency bump each change the digest; `*_test.go` and the other `cmd/` entrypoints are excluded (they do not link into the embed).
2. **Build-time injection + committed manifest** — `build-cshared.sh`. The build computes the hash via (1) and injects it into `internal/version.SourceHash` via `-ldflags -X`, so the `RUNNING .so` reports it through `mobile.Status()` (`sourceHash` field → Kotlin `ApiStatus.sourceHash`). On a successful build it also writes the committed manifest `core/apiengine/src/main/resources/api-source.hash`.
3. **CI gate** — `scripts/check-api-app-sync.sh`. Wired into `scripts/ci.sh` (both `--changed-only` and `--full`). Recomputes the live hash and compares it to the committed manifest; on mismatch it exits 1 loudly (on-device API embed is STALE vs lava-api-go — rebuild `liblavaapi.so` via `build-cshared.sh`). Pure bash + git, fast, no failure-swallow (§6.J).
4. **Gradle input-correctness** — `core/apiengine/build.gradle.kts`. The `buildCshared` task declares the lava-api-go embed-source dirs + `go.mod/go.sum` as inputs (and the manifest as an output), so Gradle re-runs `build-cshared.sh` on ANY embed-source change — killing the stale-cache drift at the root. (Before this fix the task declared only the script as an input and went up-to-date whenever the `.so` outputs existed, even after the Go source changed.)

Runtime proof on device — api-app Challenge C05.

`Challenge05ApiEmbedSourceHashMatchesTest` starts the real embed (real UI → real controller → real `NativeApiEngine` → real `.so`) and asserts the

running `embed's ApiStatus.sourceHash` equals the APK's `BuildConfig.LAVA_API_SOURCE_HASH` (both computed from the SAME script), with an empty-hash hard-fail. Equal hashes prove the packaged embed equals the source the APK was built against.

Go contract test. `lava-api-go/tests/contract/sourcehash_contract_test.go` asserts hash stability, manifest-matches-live, and falsifiability (a content edit to a tracked embed-linked `.go` changes the hash).

Per-script guides: <docs/scripts/compute-api-source-hash.sh.md> and <docs/scripts/check-api-app-sync.sh.md>.

6. Constitutional alignment (what the landed code already satisfies)

- **§6.R No-Hardcoding** — SQLite path, bind addr, port, header name, and NDK path/API level all come from config / env / Start-config, never literals.
- **§6.H Credential Security** — the auth key is surfaced only via `Status()`, never logged; plaintext UUID blobs are zeroized after hashing.
- **§6.J Anti-Bluff** — the embed reuses the production router so embed tests guarantee production behavior; the build script reports real `.so` evidence; `FakeApiEngine` reproduces the production branches (single-instance, stop-not-a-no-op, auth-always-on) rather than being a simpler stub; `ApiEngineController` is Android-free so its load-bearing test wires the REAL controller to the fake engine; Running state fields come from the real `ApiStatus` + real LAN IPs, never synthesized.
- **§6.U No sudo/su** — the build uses user-local NDK tooling only.

What Phases C + D-infra add

- **§6.R No-Hardcoding** — the controller's default port is read from `ApiConfig`'s own default (the literal lives in `:core:apiengine`), the advertised version is read from `engine.status().version`, the LAN IPs are the real interface addresses, and `:api-app`'s signing reuses `:app`'s `.env`-driven keystore inputs rather than inventing keys.
- **§6.H Credential Security** — `ApiKeyStore` holds the credential in `EncryptedSharedPreferences`, zeroizes the plaintext UUID blob after

encoding, and never logs / toStrings / exception-concatenates the key; it is surfaced only through the deliberate D-ui display path.

7. References

- Spec: [docs/superpowers/specs/2026-06-02-lava-api-android-app-design.md](#)
- Plan: [docs/superpowers/plans/2026-06-02-lava-api-android-app.md](#)
- Script guide: [docs/scripts/build-cshared.sh.md](#)
- User guide: [docs/guides/ON_DEVICE_API_USER_GUIDE.md](#)
- Discovery flow: [docs/LOCAL_NETWORK_DISCOVERY.md](#)
- Architecture overview: [docs/ARCHITECTURE.md](#)